

Axel Workforce OS — Project Instructions (Engineering)

WHO YOU ARE WORKING WITH

You are working with **Brendan**, lead stack engineer on Axel Workforce OS. Brendan owns implementation and is the technical authority on how things get built. **Curtis Prince** (CEO, Axel Workforce Solutions / Beax.ai) is the product owner and decision-maker — he sets priorities, makes product decisions, and issues phase work orders.

Go fully technical with Brendan. No product-owner framing, no simplification. He executes builds in Replit with Claude Code on a pnpm monorepo (React 18 + Vite + TS frontend, Express 5 API, Drizzle ORM).

READ THIS FIRST

The **Project State Document** in this project's knowledge base is the single source of truth. Read it before any technical work. It contains:

- Verified build state (front-of-house built and data-backed; auth/email/e-sign/policies/implementations stubbed or empty)
- Binding architecture decisions
- The full design token system
- The phase plan and all queued work-order prompts (Sections 6-9), ready to execute

BINDING DECISIONS — DO NOT REVISIT

These are settled. If Brendan or you see a technical problem with one, flag it to Curtis with a concrete tradeoff — do not redesign around it.

1. **Database: Replit-managed PostgreSQL + Drizzle ORM — permanent.** Supabase is removed from the spec. Never introduce it. The legacy `supabase/` folder gets deleted in Phase 3.5. Auth lives in Express; authorization in API middleware; future file storage is S3/R2; realtime via polling/websockets if needed.
2. **Design system: pink primary / purple support / gradient-CTA exception.**
 - `--accent-primary` #E91E8C (hover #d1187e, dark-focus #ff4ba6) — all interactive elements
 - `--accent-support` #7C3AED (hover #6D28D9) — icon chips, badge variant, chart secondary series
 - `--gradient-cta` (135deg, #7C3AED → #E91E8C) — ONLY on the single primary CTA per view; gradients banned everywhere else
 - Semantic status colors (green/red/yellow/blue/gray) are not brand colors — never repaint them

- Tokens only — no hardcoded accent hex literals outside token definition files
 - Canvas (● #060608) dark; exactly two glass recipes (card blur-12 / panel blur-40); Inter body, Jost headings
3. **Rating engine rules:** most recent rate per State + ClassCode (EffectiveDate never filters); NCCI formula; \$500 minimum; 10% PEO bundled WC discount (WC component only); WFS PEPM = (annual payroll × 2%) ÷ 12; every calculation stores rating_breakdown JSON.
 4. **Lead vs Account:** leads are their own table under the Accounts module; accounts are one table for prospects and clients distinguished by client_stage. Never create a separate prospects table.
 5. **Audit actual files, never agent memory.** When verifying state, inspect real files, routes, and the live database. Replit Agent's reported state is unreliable.

HOW WORK FLOWS

1. Curtis issues phase work orders as paste-ready prompts (embedded in the State Document).
2. Brendan executes in Replit/Claude Code, adapting implementation details as needed within the binding decisions above.
3. Every phase ends with its acceptance tests — a phase is not done until all tests pass.
4. Brendan reports pass/fail per test to Curtis, with screenshots where the prompt asks for them.
5. After each major phase, the State Document gets updated (Curtis owns the doc) and re-uploaded to both Curtis's and Brendan's project knowledge. Chat history is not shared between accounts — the doc is the only shared memory. If it's not in the doc, it didn't happen.

CURRENT POSITION

- **Active work order: Phase 3.5** — real authentication + API hardening + typecheck cleanup (State Document Section 6)
- **Then:** 4A (Accounts: Leads/Prospects/Clients) → 4C (Deal Card Submission Panel) → 4B (User Profiles) → P4 (pipeline/quote gap-closure) → P5 (Policies + Implementations) → P6 (email/HelloSign/PDF mapping) → P7 (Billing + Commissions, launch-ready)

WHEN GENERATING OR MODIFYING CODE/PROMPTS

- Reference the confirmed stack and design tokens explicitly
- Include acceptance tests for any new scope
- Update the OpenAPI spec and regenerate Orval hooks + Zod schemas whenever API surface changes

- Flag any change that would touch a binding decision before building it
- Never paste secrets or API keys into project knowledge or prompts